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Estimation of shear modulus in media with power law characteristics

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ABSTRACT

Shear wave propagation in tissue generated by the radiation force is usually modeled by either a lossless or a classical viscoelastic equation. However, experimental data shows power law behavior which is not consistent with those approaches. It is well known that fractional derivatives results in power laws, therefore a time fractional wave equation, the Caputo equation, which can be derived from the fractional Kelvin–Voigt stress and strain relation is tested. This equation is solved using the finite difference method with experimental parameters obtained from the existing literature. The equation is characterized by a fractional order which is also the power law exponent of the frequency dependent shear modulus. It is shown that for fractional order between 0 and 1, the equation gives smaller shear modulus than the classical model. The opposite situation applies for fractional order greater than 1. The numerical simulation also shows that the shear wave velocity method is only reliable for small losses. In our case, this is only for a small fractional order. Based on the published values of fractional order from other studies, there is therefore a chance for biased estimation of the shear modulus.

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1. Introduction

Shear wave elasticity imaging (SWEI) has become an important tool for characterizing tissue properties. It is done by inducing a shear wave through the radiation force of a focused ultrasonic beam [1]. Understanding shear wave propagation and the material response of human tissue can provide more accurate information for medical diagnosis, device design and treatment planning. One key aspect of this is to find equations that accurately model wave attenuation and dispersion.

However, there are some challenges for finding accurate equations for dynamic processes that occur in biological tissues. For example, it is well known that attenuation of both shear and compressional waves follows a power law [2–4]

$$a_k(\omega) = a_0\omega^y, \tag{1}$$

where the subscript k denotes that $\alpha_k(\omega)$ is the imaginary part of the wave number k, α_0 is the absorption coefficient, ω is the angular frequency, and y is the power law exponent. The classical integer order wave equations can only model attenuation with $y = 0, 2$. Therefore attenuation with $y \neq 0, 2$ is called anomalous attenuation [5].

In some cases, the stress–strain relationship is nonlinear [6]. These studies require large relative tissue deformations in the order of 20%. Here we assume very small tissue deformations so that the medium's response is linear.

There is experimental data that shows the value of the power law y for shear wave attenuation. Holm and Sinkus analyzed data published by Asbach et al. and found $y \approx 0.73$ for shear wave propagation in liver using MR elastography [3,7]. Data for malignant lesions in breast shows $y \approx 0.13$ [8]. In cell rheology, Fabry et al. [9] suggested $y \approx 1.2$ in the frequency range from 0.01 to 10 Hz. Thus there are plenty of interesting experimental results in the literature, and this paper therefore does not report any new such results.

These values of the power law are well suited for modeling with fractional calculus. It is a powerful tool for describing such attenuation in a parsimonious way, i.e. with only a few parameters. Up to now, many researchers have used fractional wave equations to model mainly compressional waves in medical ultrasound. For instance, Treeby and Cox [10] modeled power law absorption and dispersion for propagation in photoacoustic imaging using the fractional Laplacian operator. Caputo et al. [11] simulated waves in a biological medium based on the Kelvin–Voigt fractional derivative stress–strain relation. Chen et al. [12] modeled compressional wave propagation in breast using the modified Szabo's equation which has a positive fractional derivative lossy operator.

Compared with the case for compressional waves, not so many have tried to model shear waves in elastography with arbitrary power laws. But Klatt et al. [13] fitted the fractional Zener model to data for brain and liver viscoelasticity. A unifying wave equation

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for shear and compressional was discussed by Holm and Sinkus [3]. Likewise, Holm and Näsholm [4] compared several fractional wave equations and discussed the conditions under which equations were suitable for compressional waves in medical ultrasound and for shear elastography respectively.

The usual way of simulating shear waves in elastography is either to assume a lossless medium [14] or to assume a viscous medium [15]. However this does not capture some of the frequency dependency of published measurement over the last decades. The purpose of this paper is therefore to build on published data on frequency dependency, and use those values in the fractional wave equations in order to study shear wave propagation and find if a variation in the power law characteristics can lead to a bias in the estimation of e.g. the shear modulus.

In this paper, we first introduce the Caputo equation which has previously been found to be a useful model in elastography [4], and analyze the dispersion relation. Then we give a brief introduction to shear wave elasticity imaging and find representative values for the power law parameter that different studies have come up with. In the following part, we present the numerical scheme for simulating fractional wave equations, and perform a simulation study using representative values for the fractional order. We then simulate both the peak displacement method and the time-to-peak method and analyze the results.

2. Lossy wave equations

A lossy wave equation can be written as

$$\nabla^2 u - \frac{1}{c_t^2}\frac{\partial^2 u}{\partial t^2} + Lu = 0, \tag{2}$$

where c_t is the propagation velocity for the transverse wave, i.e. the shear wave, u is the displacement, and L is the loss operator. Different loss operators influence attenuation and dispersion properties, and some of them are also only valid over a restricted frequency range (usually only low frequencies). For example, $Lu = \partial u/\partial t$ gives the power law exponent $y = 0$. Another case is the Stokes wave equation, which has $Lu = \partial(\nabla^2 u)/\partial t$ and $y = 2$ for low frequencies.

2.1. Fractional wave equation

In order to model power law behavior, we build on the fact that the Fourier transform of the fractional derivative of a function, $f(t)$, generalizes the result for an integer order derivative and is $(i\omega)^\alpha F(\omega)$, where $F(\omega)$ is the Fourier transform of $f(t)$. The fractional derivative is introduced via a loss operator which can be found by starting with a constitutive equation with a fractional relation between stress T and strain S [3]

$$T = \mu S + \eta\frac{\partial^\alpha S}{\partial t^\alpha}, \tag{3}$$

where μ is the stiffness, η is the viscosity, and α is the fractional order. The range for α is between 0 and 2, but mostly it is less than 1. The classical Kelvin–Voigt model is a special case of Eq. (3) if we let $\alpha = 1$. The mechanical representation of the fractional Kelvin–Voigt model is shown in Fig. 1, where the spring represents the elastic part, parameterized by the stiffness μ, and the dashpot (fractional damper) is parameterized by the viscosity η and the fractional order α. This particular mechanical model and its generalization in the fractional Zener model have turned out to be surprisingly accurate for a wide range of materials [16].

There are several ways to define the fractional derivative. Here we use the approach of Caputo because it has physically interpretable initial conditions. Other definitions can be found in [17]. The Caputo fractional derivative is

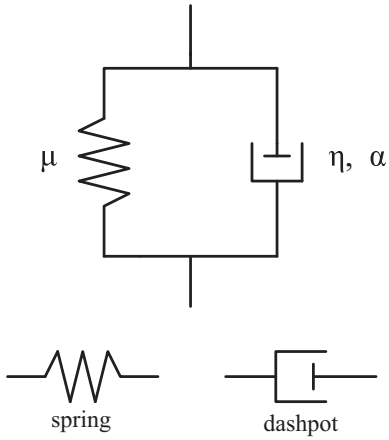


Fig. 1. The mechanical representation of the fractional Kelvin–Voigt model consisting of a Hookean spring in parallel with a fractional derivative dashpot.

$$\frac{\partial^\alpha u(t)}{\partial t^\alpha} = \frac{1}{\Gamma(n-\alpha)}\int_a^t (t-\xi)^{n-\alpha-1}\frac{d^n u(\xi)}{d\xi^n}d\xi, \qquad n-1 < \alpha < n. \tag{4}$$

Fractional calculus, allowing integrals and derivatives of any real order, is the generalization of classical calculus [17]. Fractional calculus deals with integro-differential operators and equations where the integrals are convolutions with power law type kernels. An important feature of fractional differential operator is its non-locality. In the context of Eq. (3), this means that it describes a damper with a power law memory in time.

To derive the fractional equation, we also need the relation between strain, S, and displacement, u [2]

$$S = \frac{\partial u}{\partial x}, \tag{5}$$

and the principle of conservation of momentum

$$\rho\frac{\partial^2 u}{\partial t^2} = \frac{\partial T}{\partial x}, \tag{6}$$

where ρ is the density. Combining Eqs. (3), (5) and (6), and extrapolating from 1-D to 3-D, we get

$$\nabla^2 u - \frac{1}{c_t^2}\frac{\partial^2 u}{\partial t^2} + \tau^\alpha\frac{\partial^\alpha}{\partial t^\alpha}\nabla^2 u = 0, \tag{7}$$

where the characteristic relaxation time of the medium is given by $\tau^\alpha = \eta/\mu$, and the shear wave speed is related to the elasticity by

$$\mu = \rho c_t^2. \tag{8}$$

In a lossless medium c_t is the phase velocity which is constant with frequency, while in a lossy medium, c_t is the asymptotic value of the phase velocity at zero frequency. For the convenience of the following numerical simulation, we rewrite Eq. (7) by multiplying with c_t^2

$$\frac{\partial^2 u}{\partial t^2} - c_t^2\nabla^2 u - \frac{\eta}{\rho}\frac{\partial^\alpha}{\partial t^\alpha}(\nabla^2 u) = 0, \tag{9}$$

where $\eta/\rho = c_t^2\tau^\alpha$.

It should be noted Eq. (7) was first derived by Caputo in [18] and therefore we call it the *Caputo wave equation* here. Holm and Sinkus [3] showed that Eq. (7) can be used both for shear waves and compressional waves. This was further developed in [4] where it was shown that it is a simplified version of a fractional Zener wave equation. The Caputo wave equation is valid for reasonable values of the product $\omega\tau > 1$ as long as it is not too large. This is satisfied here where the maximum value of the product will turn out to be 28.7 (Section 3). This paper is restricted to linear models

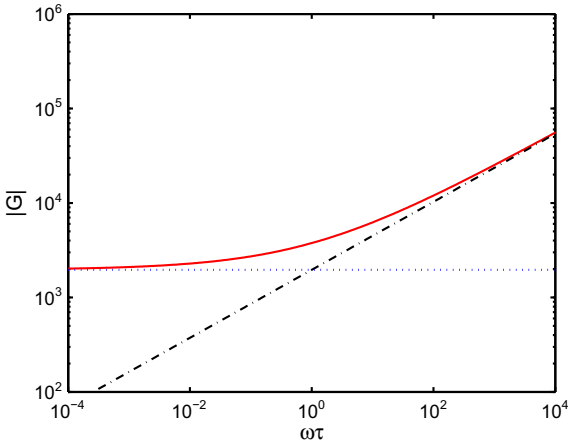


Fig. 2. The magnitude of $|G^*|$ for the liver data in Table 1 with fractional order $\alpha = 0.36$. The dot line and dash-dot lines are the approximations in the low and high regime respectively.

only. Nonlinear wave models have been discussed in e.g. [19], but only for fluids, i.e. compressional waves, and not for shear waves.

2.2. Analysis of the Caputo wave equation

We assume a plane wave solution of Eq. (7) in a complex exponential form since any general displacement field can be expressed as a sum of plane waves

$$u = \exp\ i(\omega t - kx), \tag{10}$$

where k is the wave number. We then get the dispersion relation [3]

$$k^2 - \omega^2/c_t^2 + (i\tau\omega)^\alpha k^2 = 0, \tag{11}$$

$$k^2 = \frac{\omega^2}{c_t^2}\frac{1}{1+(i\omega\tau)^\alpha} = \frac{\omega^2}{c_t^2}\frac{1}{1+(\omega\tau)^\alpha(\cos\frac{\pi\alpha}{2}+i\sin\frac{\pi\alpha}{2})}. \tag{12}$$

The attenuation and the phase velocity can be found from the imaginary and real parts of k respectively [20]

$$\alpha_k(\omega) = -k_\Im, \tag{13}$$

$$c_p = \frac{\omega}{k_\Re}. \tag{14}$$

Holm and Sinkus [3] analyzed the dispersion relation and found asymptotic results for $\omega\tau$ much less than unity, i.e. low attenuation media, and for the case where it was very large. It is generally agreed that the low $\omega\tau$ case applies to compressional waves in medical ultrasound. In that case the power law exponent of the attenuation is given as $y = \alpha + 1$. For high values of $\omega\tau$ relative to unity, the power law exponent is related to the fractional order by $y = 1 - \alpha/2$. Often the high $\omega\tau$ case is assumed for shear wave elastography, but as we will see in the data from Table 1 that may not always be the case.

2.3. Complex shear modulus of fractional order

The fractional order α is also linked to the complex shear modulus and its frequency dependent behavior. It can be found by Fourier transforming the constitutive relation of Eq. (3). The complex shear modulus is the ratio of the Fourier transforms of the stress and the strain and it is denoted by $G^* = G_d + iG_l$. The real part is the *dynamic modulus* and the imaginary part is the *loss modulus*

$$G^*(\omega) = T(\omega)/S(\omega) = \mu + \eta(i\omega)^\alpha = \mu + \eta\omega^\alpha\ \cos\frac{\pi\alpha}{2} + i\eta\omega^\alpha\ \sin\frac{\pi\alpha}{2}. \tag{15}$$

where $i^\alpha = \exp(i\alpha\pi/2) = \cos(\alpha\pi/2) + i(\sin\alpha\pi/2)$. It should be noted that there will be no frequency variation of the complex shear modulus unless there is viscosity, η. In the high $\omega\tau$ case, the first term will be negligible leading to

$$G^*(\omega) \approx \eta\omega^\alpha\ \cos\frac{\pi\alpha}{2} + i\eta\omega^\alpha\ \sin\frac{\pi\alpha}{2}. \tag{16}$$

It is evident in this case that both the real and imaginary parts of the complex shear modulus have a power law dependency of frequency with exponent α and that this behavior is asymptotically independent of the elasticity, μ.

3. Shear wave elasticity imaging

Shear Wave Elasticity Imaging (SWEI) was first described theoretically by Sarvazyan et al. [1]. It is based on the use of a shear wave remotely induced by the radiation force of a focused ultrasonic beam. Compared with other approaches in elastography, one advantage of SWEI is that the induced strain in tissue can be highly localized due to the large attenuation of the shear waves. SWEI can therefore minimize the influence of tissue boundaries.

One way for estimating shear modulus is to analyze the motion in the focal region. For example, Sarvazyan et al. [1] showed that the shear modulus could be estimated either by the maximum displacement in the focal point or by the time required for the shear wave amplitude in the focal point to reach its maximum.

The shear modulus can also be estimated by finding the shear velocity from the displacement at different lateral positions. Palmeri et al. [14] called this method the time-to-peak (TTP) method. It finds the shear modulus via Eq. (8). The TTP method is valid for purely elastic or mildly dispersive materials. Note that this method uses the shear wave speed and is different from the method of [1], where only the time required for the displacement to reach maximum at the focal point is used.

In order to find parameters for our subsequent simulations, we compiled Table 1. It lists several typical measured values for the frequency dependency of the elasticity over quite heterogenous types of tissue. All data are obtained from in vivo experiments. The sources are from SWEI [15,21], Magnetic Resonance Elastography [7,8], and Supersonic Shear Imaging [22]. Although these measurement principles are quite different, the unifying principle is that shear waves are excited in tissue and their propagation is characterized. From Table 1, we can see some results which are hard to reconcile, for example, the two different exponent α for

Table 1
Range of values of in vivo elasticity parameters and power law exponent α found from the literature and used in numerical experiments.

Material	Elasticity μ	Viscos. η	Relax. t. τ^α	Frequency Hz	Order α	Ref.
Liver	1960	7.3	0.0037	25–62.5	0.36	[3,7]
Arterial wall	4000	15.0	0.0038	0–100	1.0	[15,21]
Liver	4000	>0	>0	75–600	≈ 1.0	[22]
Malignant breast tumor	4200	33.0	0.0079	60–100	1.70	[8]

liver. This may be due to factors like differences in methods and experimental conditions. However, discussion of such factors is beyond the scope of this paper. We take all numbers in the table at face value and use them in the later numerical experiments.

To understand the frequency dependency of the shear modulus, G^*, Figs. 2 and 3 shows its behavior for the smallest and largest values of α, i.e. 0.36 and 1.7, with their low and high frequency asymptotes. It is interesting to note that the shear modulus approaches the high frequency asymptote from above for low values of α and from below for the highest values. This is because the sign of the cosine term in Eq. (15) switches sign.

The modulus μ in Table 1 is often an approximation since the value has been estimated from the lowest frequency used during the measurement, often via the zero frequency relationship between elasticity and propagation speed of Eq. (8). Several of the methods will therefore have a tendency to a slight overestimation of the elasticity μ as it is really the value at zero frequency according to Eq. (15). This comment applies to the three first entries in the Table, but not to the breast data since Eq. (15) was actually used in [8] to extrapolate the value to zero frequency. The value $\alpha \approx 1$ for liver comes from Fig. 11 in [22] since the shear wave speed there is approximately proportional to the square root of frequency. In this case, the medium was therefore found to be viscoelastic in the classical sense without the need for a fractional model.

The values of Table 1 can also be compared to rheological data such as those from Stamenović et al. [23] who found a power law dependent magnitude of the dynamic modulus for cells in the range 1–1000 Hz with $\alpha \approx 0.2$. One thing which is evident from the table is that there is no clear consensus on the value for the fractional order α. Our simulations will therefore reflect this large diversity.

It is also of interest to compute the value of the relaxation time frequency product, $\omega\tau$, for the various cases in order to check which range of the solution of the fractional wave equation one is operating in. For $\alpha = 0.36, \omega\tau \approx 5.9 \times 10^{-5}$, so it is operating in the low frequency approximation regime. For the arterial wall example with $\alpha = 1, \omega\tau$ is 2.4 at its maximum, so this case is in the cross-over region and in neither of the asymptotic regimes. For $\alpha = 1.7, \omega\tau \approx 28.7$, so it is operating in the high frequency approximation regime.

4. Simulation of shear wave propagation

In order to test the ability of the Caputo equation to predict dispersion and absorption of propagating shear waves in human tissue, we consider a practical case where the shear wave is generated by a cylindrically symmetric ultrasonic beam. All variables depend on cylindrical radius r. Eq. (9) in cylindrical coordinates can then be written as

$$\frac{\partial^2 u}{\partial t^2} - \left(c_t^2 + \frac{\eta}{\rho}\frac{\partial}{\partial t^\alpha}\right)\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) = F, \tag{17}$$

where F is the radiation force. The initial and boundary conditions are

$$u(t=0) = \frac{\partial u}{\partial t}(t=0) = 0, \tag{18}$$

$$\frac{\partial u}{\partial r}(r=0) = u(r=5\text{ cm}) = 0. \tag{19}$$

It should be noted that the fractional wave equation of Eq. (7) can be used both for the shear wave and the compressional wave that causes it. Here, we only focus on the shear wave and assume a given radiation force. This is done because we assume that the details of the compressional wave absorption is not so important for the generation of the shear wave. Therefore the force, F, is chosen to have a Gaussian shape in time and space just like in Eq. (37) in [15]

$$F = \frac{\partial Q}{\partial x}\frac{1}{\rho}M_0^2 \exp\left(-\frac{r^2}{D^2} - \frac{t^2}{T^2}\right), \tag{20}$$

where x is the axis of the cylindrical coordinate in the direction of the acoustic beam, Q determines the magnitude of shear radiation force and is $Q \approx -\rho c_l^2$, and c_l is the velocity of the longitudinal wave.

The other parameters are as follows. The characteristic beam radius is $D = 0.17$ cm, the characteristic pulse duration is $T = 0.6$ ms, the density is $\rho = 1000\,\text{kg/m}^3$, the Mach number $M_0 = 0.001$, and the longitudinal wave velocity is $c_l = 1500$ m/s. We assume that $\partial Q/\partial x$ varies by a factor 0.2 at a scale of 0.5 cm. The values of all these parameters are obtained from [15].

4.1. Numerical strategy

To establish the numerical scheme, we let the grid size in time and space be Δt and Δr respectively. The grid points in time are $t_n = n\Delta t, n = 0, 1, \ldots, N_t$ and the grid points in space are $r_k = k\Delta r, k = 0, 1, \ldots, N_r$.

The finite difference approximation for the second order spatial derivative is

$$\frac{\partial^2 u(r_k, t_n)}{\partial t^2} \approx \frac{u_k^{n+1} - 2u_k^n + u_k^{n-1}}{(\Delta t)^2}, \tag{21}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u(r_k, t_n)}{\partial r}\right) \approx r_{k+\frac{1}{2}}\frac{u_{k+1}^n - u_k^n}{r_k \Delta r^2} - r_{k-\frac{1}{2}}\frac{u_k^n - u_{k-1}^n}{r_k \Delta r^2}. \tag{22}$$

Let $z = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)$, the approximation of the fractional derivative, be divided into two cases just as in [24]

- $0 < \alpha < 1$:

$$\frac{\partial^\alpha z(r, t_{n+1})}{\partial t^\alpha} = \frac{1}{\Gamma(1-\alpha)}\sum_{j=0}^{n}\int_{t_j}^{t_{j+1}} (t_{n+1} - \xi)^{-\alpha}\frac{\partial z(r,\xi)}{\partial \xi}d\xi \approx \frac{(\Delta t)^{1-\alpha}}{\Gamma(2-\alpha)}\sum_{j=0}^{n} v_{n-j}\left[\frac{z(r, t_{j+1}) - z(r, t_j)}{\Delta t}\right], \tag{23}$$

- $1 < \alpha < 2$:

$$\frac{\partial^\alpha z(r, t_{n+1})}{\partial t^\alpha} = \frac{1}{\Gamma(2-\alpha)}\sum_{j=0}^{n}\int_{t_j}^{t_{j+1}} (t_{n+1} - \xi)^{1-\alpha}\frac{\partial^2 z(r,\xi)}{\partial \xi^2}d\xi \approx \frac{(\Delta t)^{2-\alpha}}{\Gamma(3-\alpha)}\sum_{j=0}^{n} w_{n-j}\left[\frac{z(r, t_{j+1}) - 2z(r, t_j) + z(r, t_{j-1})}{(\Delta t)^2}\right], \tag{24}$$

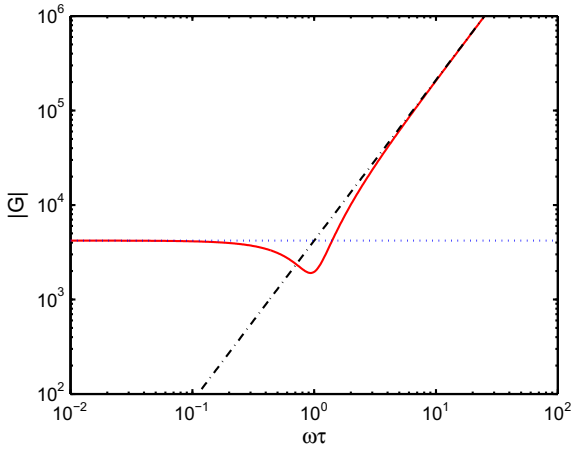


Fig. 3. The magnitude of $|G^*|$ for the breast data in Table 1 with fractional order $\alpha = 1.7$. The dot line and dash-dot lines are the approximations in the low and high regime respectively.

where $v_{n-j} = (n-j+1)^{1-\alpha} - (n-j)^{1-\alpha}, w_{n-j} = (n-j+1)^{2-\alpha} - (n-j)^{2-\alpha}$. Eq. (17) can be solved numerically by inserting Eqs. (21)–(23) (or Eq. (24) for $\alpha > 1$) into Eq. (17). Details about the numerical method can be found in [24,25].

Solving fractional differential equations is a time-consuming task due to the non-local property of the fractional differential operator. We used Matlab R2013a to solve Eq. (17) numerically using Intel Xeon processor L5640, and the operating system was Linux. Assuming $N_t = 1500$ and $N_r = 100$, the simulation time for integer order $\alpha = 1$ is 0.41 s while for $\alpha = 0.36$, 1.7, the time consumption is 41.7 and 58.2 s, respectively, i.e. an increase by more than a factor of 100.

5. Results and discussion

In the first example, the shear displacement in the focal point, $r = 0$, is found. We took the fractional order from Table 1 as $\alpha = 0.36$, 1 and 1.7, but let the values for shear modulus and the viscosity be the same for all cases in order to ease the comparison. We used the values $\mu = 4000$ Pa and $\eta = 15$ Pa s. The value for the shear modulus corresponds to a shear speed of $c_t = 2$ m/s, which is the asymptotic value as the frequency approaches zero.

In the numerical simulation, we used $\Delta t = 10^{-5}$ s, $\Delta r = 5 \times 10^{-4}$ m. These values were chosen based on the stability of the non-fractional wave equation, Eq. (12) in [12], where they said that for a fractional wave equation, an even stricter condition should be satisfied. We followed this criterion in our simulations. With these parameters, we were not able to detect any signs of instability in our solutions. We also verified that the solution of the fractional equation for values very close to an integer, e.g. $\alpha = 0.99$, was similar to the solution of the classical wave equation $\alpha = 1$ as further evidence of a stable solution.

The shear wave displacement at the focal point is shown in Fig. 4. We can see that the temporal behavior at the focal point varies greatly with α. The peak amplitude is higher and the wave attenuates faster for small α. For the classical model $\alpha = 1$, the amplitude at the focal point can reach around 7 μm. This is our reference. For $\alpha = 0.36$, the amplitude is 3 times larger and occurs earlier. For $\alpha = 1.7$, the amplitude only reaches up to 20% of the value for the non-fractional case and the peak does not occur until around 42 ms (which is not shown in the figure). For the lossless equation, $\eta = 0$, the amplitude is only approximately 4% higher than the case $\alpha = 0.36$, so the plots are similar.

In order to verify that our pulses were in the range of frequencies reported in Table 1, we also did a spectrum analysis. It showed that the main part of the energy of the spectrum was in the range up to 70 Hz for $\alpha = 0.36$, and up to 40 Hz for $\alpha = 1$ and 1.7 at the −6 dB point.

It should be noted that the parameters in our simulation that determine the radiation force F in Eq. (17) are obtained from [15]. Therefore, the absolute values of the displacements are calibrated to the extent that values of these parameters are accurate. In our study, this does not matter so much as the relative values are more important than the absolute values of displacement.

Thus lossless media or those with a small value of α give the highest amplitude of the displacement at the focal point. There is a qualitative method for comparing various tissues based on the peak displacement [1]. It builds on the lossless version of Eq. (3) which in combination with Eq. (5), says that the strain, S, and therefore also the displacement, u, both are inversely proportional to the shear modulus for a given stress, T. Our example shows that this method is also sensitive to changes in α and η.

In order to give an example of this, we varied the elasticity in the simulation for $\alpha = 0.36$ until the peak value matched that of the case for $\alpha = 1$ in Fig. 4. The value of μ for $\alpha = 0.36$ had to be increased from 4000 Pa to 25,000 Pa (c_t increased from 2 to 5 m/s) in order to get approximately the same peak amplitude, see Fig. 5. So in this particular example, assessing the shear modulus with the peak displacement method can give widely different results depending on the loss and fractional order.

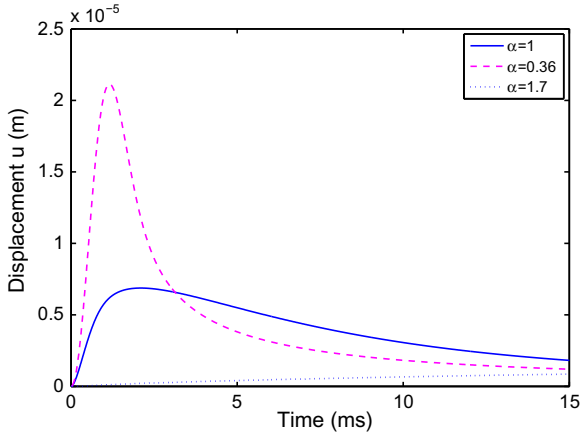


Fig. 4. Comparison of shear displacement at $r = 0$ for $\alpha = 0.36$, 1, and 1.7. The viscosity $\eta = 15$ Pa s and wave speed $c_t = 2$ m/s.

We also simulated a propagating shear wave in order to estimate the propagation speed and reconstruct the shear modulus based on the time-to-peak (TTP) method of Palmeri et al. [14]. We wanted to see how the result was affected by the fractional order, α, in the Caputo equation. In the next numerical experiment, we let the lateral positions be $r = 0$, 10, 15, 20, and 25 mm. Figs. 6–8 show the simulation results for different values of α. Note that both the time axis and the amplitude scale of the three figures are different.

For $\alpha = 0.36$ in Fig. (6), the peaks at the different lateral locations are distinct. The averaged shear wave speed is 2.00 m/s which is also the true value. The averaging takes place by finding a velocity estimated from each pair of neighbor peaks and then averaging the four values. Thus, the TTP method is not affected when α is small and the medium is close to lossless. When $\alpha = 1$, Fig. 7, and $\alpha = 1.7$, Fig. 8, it is much harder to separate the peaks. The averaged shear wave speeds are 1.74 m/s and 8.94 m/s respectively. For these two cases, the TTP method is highly biased, most likely because it is only valid for purely elastic or mildly dispersive materials, i.e. media with value of $\omega\tau \ll 1$. As shown before, this is not the case for $\alpha = 1$ and 1.7, where the loss term has significant effect.

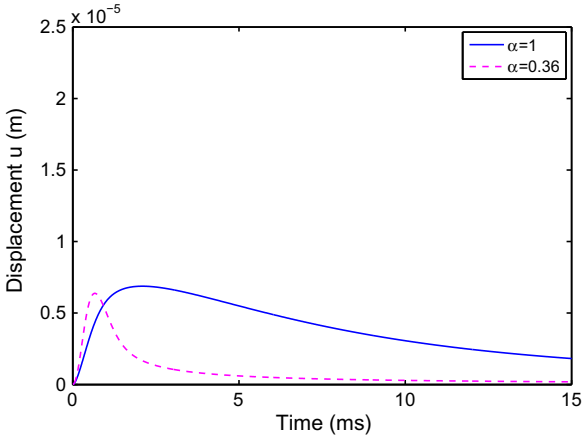


Fig. 5. Comparison of shear displacement at $r = 0$ for $\alpha = 0.36, \eta = 15$ Pa s, $\mu = 25$ kPa and $\alpha = 1, \eta = 15$ Pa s, $\mu = 4$ kPa.

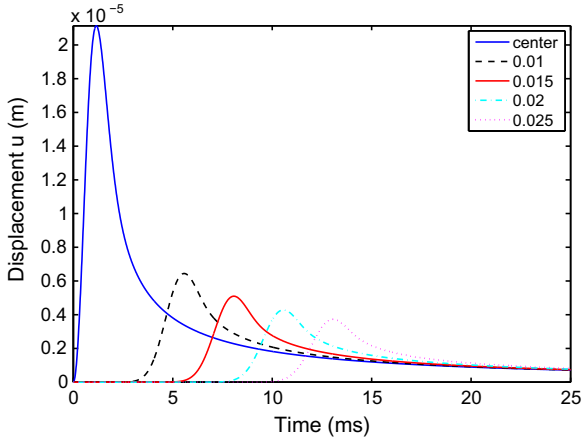


Fig. 6. Displacement at different lateral locations: $r = 0$, 10, 15, 20, 25 mm for $\alpha = 0.36$.

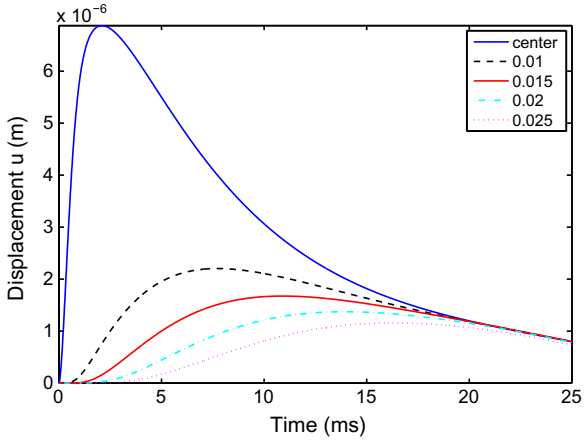


Fig. 7. Displacement at different lateral locations: $r = 0$, 10, 15, 20, 25 mm for $\alpha = 1$.

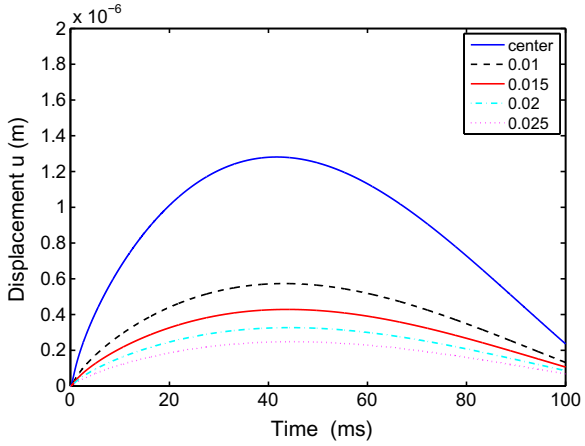


Fig. 8. Displacement at different lateral locations: $r = 0$, 10, 15, 20, 25 mm for $\alpha = 1.7$.

According to [3,26], the phase velocity increases with frequency, so we expect an estimation of the shear velocity of $c_t = 2$ m/s or more. In our opinion the value of 8.94 m/s may be too high and the value of 1.74 m/s is definitely too low. This is most likely because the TTP method is only valid as long as the pulse is well concentrated in time, i. e. in dispersionless or lossless media.

Finally, let us comment on a result by Sarvazyan et al. [1]. They gave a theoretical model for SWEI, and showed a comparison of simulation and measurement of the time evolution of the shear displacement along the radial position in their Fig. 13. From their figure, we can see that at the focal point, the simulated maximum amplitude is less than 2 μm while the experimental data shows an amplitude around 10 μm. Based on our results, one possible reason for this discrepancy could be the wave equation they used (Eq. (3) in [1]) which is the viscoelastic equation of integer order ($\alpha = 1$). If however, their measurement came from a medium with $\alpha < 1$, the amplitude would have been larger than they predicted just as the measured data.

6. Conclusion

A time fractional wave equation, the Caputo wave equation, is used to simulate shear wave propagation in viscoelastic media generated by a focused ultrasonic beam. This equation can be derived from the fractional Kelvin–Voigt stress–strain relation and accounts for power law attenuation. We used experimental data and found $\alpha = 0.36$, 1, and 1.7 from existing literature as representative values.

We then studied the relation between fractional order and the displacement at the focal point. Simulation results demonstrate that a higher value of the fractional order gives lower displacement which means a higher estimate of the shear modulus. Therefore, the fractional order is important to fully characterize the viscoelasticity of media.

The shear wave speed method is also used in the simulation to estimate elasticity. The results show that the estimation for lossless media or media with small losses works well with this method. Among our examples, only the case for a small fractional order has small losses. This means that in the simulation, the examples with $\alpha = 1$, and 1.7 do not give reliable results with this method.

If the published values of viscosity and fractional order are reliable, there is therefore a chance for biased estimation of the shear modulus when the principle of acoustic radiation force excitation is used. On the other hand, our work may also support the hypothesis that the reported values are inconsistent due to, for instance, shortcomings in the experimental methods.

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